

RetroDiff: Training Objective Derivation

Let x denote a clean data sample and x_t its noised counterpart at diffusion step $t \in \{1, \dots, T\}$. The forward process $q(x_{1:T}|x)$ gradually corrupts x ; the model p_θ learns to reverse it. Applying Jensen’s inequality to the log-likelihood yields the *evidence lower bound* (ELBO):

$$\log p_\theta(x) \geq \mathbb{E}_{q(x_{1:T}|x)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x)} \right] = -\mathcal{L}_{\text{ELBO}} \quad (1)$$

The bound in (1) is tight when $q = p_\theta$. Expanding the joint by the Markov structure of the diffusion chain and applying Bayes’ theorem to each reverse step decomposes $\mathcal{L}_{\text{ELBO}}$ into a reconstruction term and a sum of per-step KL divergences:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x_0|x_1)]}_{\text{reconstruction}} + \sum_{t=2}^T \underbrace{\text{KL}(q(x_{t-1}|x_t, x) \parallel p_\theta(x_{t-1}|x_t))}_{\text{step-}t \text{ denoising}} \quad (2)$$

Each KL term in (2) is tractable because the forward posterior $q(x_{t-1}|x_t, x)$ is Gaussian. **RetroDiff**’s key contribution is to condition the entire objective on a set $R(x)$ of k exemplars retrieved from the training corpus based on x . Averaging over the data distribution gives the final training objective:

$$\mathcal{L}_{\text{RetroDiff}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\mathcal{L}_{\text{ELBO}}(x \mid R(x))] \quad (3)$$

Equation (3) reduces to the standard DDPM objective when $R(x) = \emptyset$, making RetroDiff a strict generalisation of the baseline diffusion framework.